

THE TECHNO FIX

AN ESTIMATED 800 DEEP-TECH INDIAN VENTURES HAVE RAISED AROUND \$3.6 BILLION IN THE PAST 10 YEARS, INTEGRATING ADVANCED SCIENCE AND ENGINEERING TO IMPLEMENT CLIMATE CHANGE SOLUTIONS ACROSS SECTORS

BY PALAK AGARWAL



CLIMATE CHANGE AND its potentially calamitous consequences for Earth have compelled

a raft of Indian start-ups to seek technology-based solutions to lower toxic carbon emissions.

That these companies work in sectors ranging from energy, supply chains and mobility to advanced materials, biotech and food and agriculture—massive, high-growth segments of the economy—have made them a compelling story for investors. India has committed to net zero carbon emissions by 2070.

The gap between ambition and outcome is fuelling the new wave of climate tech start-ups. Investors aren't just backing green ideas, they're chasing deep-tech innovations across sectors to solve complex climate problems.

An estimated 800 deep-tech Indian ventures have raised around \$3.6 billion in the past 10 years (2014-2024) and are transcending traditional climate tech boundaries, integrating advanced science and engineering to drive climate change solutions across sectors.

Take, for instance, Varaha Earth, which has made a measur-

able impact since its founding barely three years ago. It has helped eliminate more than one million tonnes of carbon dioxide emissions from 1.1 million acres owned by small farmers.

SCALABLE SOLUTIONS

Varaha works with 100,000+ farmers in South Asia and Africa and seeks to decarbonise farming through regenerative agriculture, afforestation and reforestation, biochar production and rock weathering or erosion of rocks and minerals on the earth's surface.

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The next growth wave lies in start-ups with disruptive IPs in fuel cells, energy storage and circular economy tech



Investor focus is moving beyond ESG mandates to measurable returns like performance, scalability and cost-efficiency



Funding models now align with both commercial and non-commercial climate ventures—enabling broader experimentation and impact